MA1522 Linear Algebra for Computing

Lesson 1: Linear Systems, Row Operations and Row-Echelon Forms

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Basic rules:

- Please (mainly) ask the questions related to the course.

Section 1.1 Linear Equations

A **linear equation** in the variables $x_1, \ldots, x_n$ is

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

with constant **coefficients** $a_1, \ldots, a_n$ and constant term b. It is **homogeneous** when $b = 0$.

Every variable appears to the first power only: no x_i^2, x_ix_j, $1/x_i$, $\sqrt{x_i}$, $\sin x_i$, ...

Example 1

Which of these are linear equations? If so, write it in standard form and say whether it is homogeneous.

1. $z - 2x - 2y = 7$
2. $y = x + 1$
3. $x(y + z) = 3$
4. $2x = 3y$
5. $z = 2(x + y)$
6. $y = (2x + 3)^2$

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1. Linear: $-2x - 2y + z = 7$. Not homogeneous.
2. Linear: $-x + y = 1$. Not homogeneous.
3. **Not** linear: xy and xz have degree 2.
4. Linear: $2x - 3y = 0$. Homogeneous.
5. Linear: $2x + 2y - z = 0$. Homogeneous.
6. **Not** linear: expanding gives $4x^2$.

Section 1.1 Linear Systems

A **linear system** is a set of m linear equations in the same n variables. It is encoded by its **augmented matrix**:

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases} \longleftrightarrow \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right)$$

- A **solution** satisfies every equation at once; the system is **consistent** if at least one exists, and the **general solution** describes all of them.
- The system is **homogeneous** if $b_1 = \cdots = b_m = 0$. A homogeneous system is always consistent: $\mathbf{0}$ is the **trivial solution**.
- Exactly one of three things happens: no solution, exactly one solution, or infinitely many.

Question 1

Consider the linear system $\begin{cases} x + 2y - z = 1 \\ \qquad y - z = 3 \end{cases}$.

Explain why

$$x = -5 - t, \quad y = 3 + t, \quad z = t, \qquad t \in \mathbb{R}$$

is a general solution of this system.

Question 1

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There are **two** things to check: every listed triple is a solution, and every solution appears in the list.

Question 1 — solution

Each triple is a solution: for every t,

$$(-5-t)+2(3+t)-t=-5-t+6+2t-t=1, \qquad (3+t)-t=3. \checkmark$$

Each solution is listed: let (x,y,z) satisfy both equations, and take $t=z$. The second equation gives $y=3+z=3+t$, and the first then gives
$x=1-2y+z=1-2(3+t)+t=-5-t$. □

Challenge

Explain why

$$x = -2 - s, \quad y = s, \quad z = s - 3, \qquad s \in \mathbb{R}$$

is *also* a general solution of the same system $\begin{cases} x + 2y - z = 1 \\ y - z = 3 \end{cases}$.

Remark

The form in which a general solution is written is not unique. The same set of solutions can be described by different parametrisations, so two answers that look nothing alike may both be correct.

Section 1.2 Elementary Row Operations

The three **elementary row operations** on a matrix:

- **Interchange** two rows: $R_i \leftrightarrow R_j \quad (i \neq j)$.
- **Multiply** a row by a nonzero constant: $aR_i \quad (a \neq 0)$.
- **Add** a multiple of a row to another: $R_i + cR_j \quad (i \neq j)$.

Every elementary row operation is **reversible**.

Section 1.2 Row Equivalence

A and B are **row equivalent** if one is obtained from the other by a series of elementary row operations,

$$A = A_0 \to A_1 \to \cdots \to A_k = B.$$

By reversibility the relation is symmetric.

Theorem

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

This licenses the whole method: replace a system by a simpler row equivalent one, without losing or gaining any solution.

Question 2

Write the system as an augmented matrix, carry out the given row operations, and read off the solution.

$$\begin{cases} 2x + 3y = 12 \\ x + 2y = 7 \end{cases} \qquad R_1 \leftrightarrow R_2 \ / \ R_2 - 2R_1 \ / \ -R_2.$$

Question 2

Write the system as an augmented matrix, carry out the given row operations, and read off the solution.

$$\begin{cases} 2x + 3y = 12 \\ x + 2y = 7 \end{cases} \qquad R_1 \leftrightarrow R_2 \;\; / \;\; R_2 - 2R_1 \;\; / \;\; -R_2.$$

$$\left(\begin{array}{cc|c} 2 & 3 & 12 \\ 1 & 2 & 7 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 2 & 7 \\ 2 & 3 & 12 \end{array}\right) \xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & -1 & -2 \end{array}\right) \xrightarrow{-R_2} \left(\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 2 \end{array}\right)$$

The last matrix reads $x + 2y = 7$ and $y = 2$, so $y = 2$ and $x = 7 - 4 = 3$.

Check in the original system: $2(3) + 3(2) = 12$ ✓ and $3 + 2(2) = 7$ ✓. □

Section 1.3 Row-Echelon Forms

A **zero row** has all entries 0. The **leading entry** of a nonzero row is its leftmost nonzero entry.

A matrix is in **row-echelon form** (REF) if

1. all zero rows are grouped at the bottom;
2. each leading entry is strictly to the right of the leading entry of the row above it.

It is in **reduced row-echelon form** (RREF) if moreover

3. every leading entry equals 1;
4. every column containing a leading entry has all its other entries 0.

A column containing a leading entry is a **pivot column**; the others are **non-pivot columns**.

A matrix has *many* REFs but exactly *one* RREF — hence "*a* REF of A" but "*the* RREF of A".

Example 2

For each matrix, state whether it is in REF, in RREF, or neither.

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$ (c) $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ (e) $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

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(a) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$ (c) $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ (e) $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

(a) **REF only:** the leading entry 4 is not 1, and pivot column 3 also contains a 3.

(b) **Neither:** the leading entry of row 2 lies to the left of that of row 1.

(c) **Neither:** the zero row is not at the bottom.

(d) **RREF:** pivot columns 1 and 2; column 3 is non-pivot.

(e) **REF only:** the leading entries 3 and 5 are not 1. A leading entry need not be 1, and an all-zero column is allowed.

Section 1.3 Reading solutions off a REF

Let R be a REF of the augmented matrix of a system in n variables.

- If the rightmost column of R is a pivot column, some row reads $0 = b$ with $b \neq 0$: the system is **inconsistent**.
- Otherwise the system is **consistent**:
  - non-pivot columns $\rightarrow$ arbitrary **parameters**;
  - pivot columns $\rightarrow$ solved by **back substitution**, from the bottom row upwards.

Number of parameters = number of non-pivot columns among the first n columns. The solution is unique exactly when all n are pivot.

Question 3

The matrix

$$\left(\begin{array}{ccc|c} 2 & -1 & 5 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

is in REF but not in RREF.

Transform it into its RREF using elementary row operations, and solve the corresponding system.

Question 3 — solution

The leading entry 2 is not 1, and pivot column 2 contains -1 in row 1.

$$\left(\begin{array}{ccc|c} 2 & -1 & 5 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1+R_2} \left(\begin{array}{ccc|c} 2 & 0 & 4 & 6 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{\frac{1}{2}R_1} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Pivot columns 1 and 2; column 3 is non-pivot, so set $z = t$. The rows read $x + 2z = 3$ and $y - z = 2$, giving directly

$$x = 3 - 2t, \quad y = 2 + t, \quad z = t, \quad t \in \mathbb{R}.$$

In RREF no back substitution is needed — each pivot variable is already isolated.

Next lessons

- **Section 1.4 Gaussian Elimination**
- **Section 1.5 More on Linear Systems**